

Question 1:

House Price Prediction using Linear Regression

Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("house_price.csv")

print("First 5 Rows:")
print(data.head())

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("\nPredicted House Prices:")
print(y_pred)

# Accuracy
score = r2_score(y_test, y_pred)
print("\nR2 Accuracy Score =", score)
```

Question 2:

Student Performance Prediction using Multiple Linear Regression

Code:-

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
data = pd.read_csv("student_performance.csv")

print("First 5 Rows:")
print(data.head())
X = data[['StudyHours', 'Attendance', 'InternalMarks', 'AssignmentScore']]
y = data['FinalMarks']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("\nPredicted Final Marks:")
```



```
print(y_pred)
score = r2_score(y_test, y_pred)

print("\nR2 Accuracy Score =", score)
```